

Tien Dung Vu

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PROFESSIONAL SUMMARY

Final-year Bachelor of Artificial Intelligence student at UTS (WAM 85.0, graduating October 2026) with hands-on experience building end-to-end data pipelines, machine learning models, and NLP analytics systems grounded in real-world data. Designed a full-stack aviation analytics platform with an XGBoost demand forecasting model ($R^2=0.952$) and Monte Carlo risk simulation, and led a commissioned three-model NLP sentiment pipeline (VADER, RoBERTa, FinBERT) for The Sydney Morning Herald analysing 1,033 articles across 10 years. Built 4 content automation pipelines and a cross-channel analytics dashboard in current role. Strong in ETL pipeline engineering, statistical modelling, and cross-functional delivery - ready to bring data science and AI capability to the AFL Tech, Data & AI team.

CORE COMPETENCIES

Data Science & Analytics

Machine Learning

NLP & Text Classification

ETL Pipeline Engineering

Python & scikit-learn

Statistical Modelling

Data Visualization

LLM & Agentic Systems

WORK EXPERIENCE

Affinity Marketing Australia May 2026 - Present

AI-Driven Marketing Intern - Sydney, Australia

- Delivered a full **SEO baseline audit** across the top 10 landing pages using Google Search Console, producing keyword ranking reports, GA4 baseline traffic analysis, and a monthly rank tracking cadence
- Built a **Looker Studio cross-channel analytics dashboard** consolidating Google Analytics 4 and Search Console data into a single performance view with automated weekly stakeholder summaries
- Engineered 4 end-to-end **content automation pipelines** using Claude Code and n8n (SEO blog expander, video-to-content-pack, blog-to-social repurposing, ad copy refresh), eliminating manual production steps across recurring campaign cycles
- Built an email subject-line **A/B testing pipeline** and a weekly analytics summary emailer, automating recurring reporting and campaign testing workflows
- Authored an **AI automation playbook** documenting all pipelines, tools (Claude Code, n8n, GA4, Looker Studio, GSC), and workflows for team handover

CMC Corporation Vietnam Oct 2024 - Apr 2025

AI Engineer Intern - Vietnam

- Built **computer vision classification pipelines** using CNNs with data augmentation and normalisation techniques to improve model generalisation across image datasets
- Developed end-to-end **deep learning pipelines** covering data preparation, training, evaluation, and hyperparameter tuning; evaluated model performance and iteratively improved architectures
- Implemented solutions in **Python** with structured experiment tracking and documented handover

FPT Vietnam Sep 2023 - Feb 2024

AI Engineer Intern - Vietnam

- Applied **supervised learning algorithms** (Linear Regression, Logistic Regression, Decision Trees) and clustering methods (K-Means) for pattern discovery across structured datasets
- Developed **NLP text classification models** using word embeddings and text-processing techniques; conducted data analysis, visualisation, and experimentation in Google Colab
- Evaluated and interpreted **machine learning results** to support data-driven decision-making

OACIS - Sydney Airport Oct 2025 - Present

Passenger Service Agent - Sydney, Australia

- Managed passenger flow and resolved enquiries across a high-volume airport environment, developing strong stakeholder communication and operational problem-solving skills

- Collaborated with airline staff, security personnel, and operations teams to maintain smooth daily workflows under time-critical conditions

PROJECTS

Pacific Wings Airline Strategy Simulator

Python, FastAPI, Next.js, XGBoost, scikit-learn, LangGraph, Gemini AI, PostgreSQL, Docker

Built a full-stack aviation analytics and strategy platform grounded in real-world data: engineered an **ETL pipeline** ingesting BITRE aviation statistics, World Bank GDP/population/tourism indicators, OurAirports coordinates, and live competitor schedules - recalibrated a market-share logit model against real benchmarks (matched Singapore Airlines' ~60% Sydney-Singapore share within 1pp)

Trained and validated an **XGBoost demand forecasting model** ($R^2=0.952$, MAPE=15.3% on held-out 2024 test set) with 5-fold cross-validation and a custom **statistical confidence-scoring system** combining bootstrap-ensemble disagreement, per-route reliability, and extrapolation distance into a single explainable 0-100 score

Built a **Monte Carlo risk simulator** sampling real fuel-price and GDP-growth volatility to produce profit/passenger/load-factor probability distributions for any route or scenario

Implemented a **LangGraph five-agent pipeline** and an AI executive chat agent backed by Gemini with 11 function-calling tools; designed a PostgreSQL schema, Docker Compose deployment, and an 11-page Next.js analytics dashboard

Australian Media Sentiment and Framing of AI Commissioned - Sydney Morning Herald

Python, VADER, RoBERTa, FinBERT, BERTopic, Guardian API | UTS Capstone

Built a **Python ETL pipeline** collecting 2,622 Guardian Australia articles via API with keyword search, rate-limited pagination, and multi-criteria AI relevance filtering, reducing the corpus to 1,033 articles spanning 2017-2026

Applied a **three-model NLP sentiment framework** (VADER, RoBERTa, FinBERT) with FinBERT sliding window (450-token chunks, 50-token stride); quantified inter-model agreement and identified a persistent headline-body sentiment divergence across all 10 major Guardian sections

Applied **BERTopic topic modelling** to identify dominant risk themes; delivered 6 evidence-based editorial recommendations to SMH; identified a structural sentiment shift post-ChatGPT (FinBERT headline compound declined from -0.110 in 2017 to -0.330 in 2026)

Misinformation Detection System

Python, scikit-learn, TF-IDF, Naive Bayes, NLP | UTS - Team of 5

Led a team of 5 to build a **binary fake-news classifier**; implemented a full NLP preprocessing pipeline with parallelised text cleaning using `joblib.Parallel(n_jobs=-1)` across all CPU cores

Vectorised text with **TF-IDF** (40,000 features, $\text{max_df}=0.7$) and trained a Multinomial Naive Bayes classifier; persisted trained model and vectoriser as .pkl files; shipped an interactive **CLI classification loop** for real-time prediction

IMDB Movie Review Sentiment Classification

Python, scikit-learn, NLTK, TF-IDF, Logistic Regression | UTS

Built a binary sentiment classifier on 50k IMDB reviews using TF-IDF (unigram+bigram, 10k features) and Logistic Regression; tuned hyperparameters via **GridSearchCV 5-fold cross-validation** achieving $F1=0.8799$ (± 0.0073); test set AUC-ROC=0.9531, AUC-PR=0.9567

EDUCATION

Bachelor of Artificial Intelligence - University of Technology Sydney

Graduating Oct 2026

WAM: 85.0 / 100

SKILLS

Programming: Python, SQL, Java, Swift

Machine Learning & AI: NLP, Computer Vision, CNNs, TF-IDF, Naive Bayes, Logistic Regression, XGBoost, scikit-learn, Text Classification, Sentiment Analysis, Topic Modelling (BERTopic)

LLM Systems: LangGraph, LLM Pipeline Design, Agentic Workflow Design, Prompt Engineering, RAG, Information Retrieval

Data & Analytics: ETL Pipeline Engineering, Data Analysis, Feature Engineering, Statistical Analysis, Data Visualization, Google Analytics 4, Looker Studio

Software Engineering: FastAPI, API Integration, Backend Development, Docker, PostgreSQL, Git, GitHub, REST APIs

Collaboration: Team Leadership, Agile Delivery, Cross-functional Communication, Technical Documentation